23.0 TEST SPECIMEN ASSEMBLY

The test specimens were all fabricated from 7075 aluminum. Therefore, the various fabrication and assembly techniques employed were those which are applicable to aluminum construction. Fabrication of the end closure test specimens is presented in Section 22.3. Details of the assembly of the local buckling specimens and the full size 40 x 40 inch panels are presented below.

23.1 Panel Test Specimens

The basic fabrication sequence applied to the full size 40 x 40 inch panel assemblies was as follows:

1) Shear to develop flat pattern width and length dimensions,

2) Numerical Control (NC) punch 0.125 inch diameter index holes (for locating on forming dies) and rectangular slots for fin and/or secondary bead trim,

3) Clean and solution heat-treat; hold in "W" condition at reduced temperature,

4) Roller flatten,

5) Brake form uniform cross-sections,

6) Reform end closures,

7) Degrease and artifically age to -T6 condition,

8) Fit-up for bonding (doublers and panels) and drill index holes for final bond cycle,

9) Clean for bonding (alkaline clean and deoxidize with sodium dichromate/sulphuric acid solution),